

cce313-short-prev-cut

Sobuj Sir

1. Chapter 1 (Basics)

Key Concepts: Network types, topology, protocols, socket, host vs end system, ISP, ARP, NAT, etc.

Question / Concept	Stars	Importance
Distinguish between host and end system; relation between hosts, clients, servers	★★	🔥 High
What is a socket address?	★	Medium
Compare bus, ring, star, mesh topologies	★★	Medium
Define unicast, multicast, broadcast	★★★★	🔥🔥 Very High
Explain store-and-forward packet switching	★★	High
Connection-oriented vs connectionless services	★★	Medium
What is congestion control, flow control, and error control	★★	Medium
Explain duties/services of OSI layers	★★★★★	🔥🔥 Very High
Four levels of addressing in TCP/IP architecture	★★	High

2. IP Address (IPv4)

Key Concepts: Subnetting, CIDR, IP class, mask, address range, subnet design

Question / Concept	Stars	Importance
What is subnetting / supernetting / CIDR and difference from default mask	★★★★	🔥🔥 Very High
Given IP → find netid, hostid, mask	★★★★	🔥🔥 Very High
Why subnet masks are used in networking	★★	High
Given IP → find first & last address, total addresses	★★★★★	🔥🔥🔥 Top Priority
Design subnets or subblocks for ISP/cities/customers	★★★★★	🔥🔥🔥 Top Priority
Design given number of subnets (e.g., 8, 122)	★★	High

Question / Concept	Stars	Importance
Calculate broadcast and network address for given mask	★★★★	🔥🔥 Very High
Determine mask bits required for given subnets	★★	High
Explain CIDR rules and restrictions	★★★★	🔥🔥 Very High
Compare subnetting and supernetting	★★	High
What is NAT and how it helps in address depletion?	★★	🔥 High
Describe three steps for subnet operation	★★★★	High

✅ Sobuj Sir — IP Address Focus Points:

1. **Subnetting calculation practice** (design & range)
2. **CIDR design** and allocation examples
3. **Subnet mask / broadcast address**
4. **Real-world ISP block division** (e.g., 150.80.0.0/16, 254.103.0.0/15)
5. **CIDR restriction questions**

3. Network Layer: Data Plane & Control Plane

Key Concepts: Routing, forwarding, algorithms (Dijkstra, distance vector), congestion

Question / Concept	Stars	Importance
Explain network layer data plane and control plane	★★	🔥🔥 Very High
Explain forwarding process in router R1 for given dest	★★★★	🔥🔥 Very High
Compare centralized vs distributed routing	★★★★	🔥🔥 Very High
Dijkstra's shortest path algorithm (with table)	★★★★★	🔥🔥🔥 Top Priority
Distance vector routing working	★★	High
Construct forwarding table / least-cost tree	★★	High
Congestion control, flow control concepts	★★	Medium
Explain datagram vs virtual circuit subnet	★★	Medium
Protocols involved in traceroute	★★	Medium

Sarna Mam

1. Application Layer

Key Concepts: HTTP, Web caching, cookies, persistent connection, MIME, etc.

Question / Concept	Stars	Importance
HTTP persistent vs non-persistent connection	★★★★	🔥🔥 Very High
Explain cookie mechanism (Alice–Amazon)	★★	High
How does web caching reduce delay	★★★★	🔥🔥 Very High
Discuss major protocols in networking (TCP, UDP, IP, HTTP, etc.)	★★	Medium
How proxy server reduces delay	★★	High
How SMTP marks end of message, compare with HTTP	★★	🔥🔥 Very High
Persistent connection close in HTTP/1.1	★	Medium
Encryption in HTTP	★	Medium
Conditional GET	★	Medium
Explain DNS, HTTP, FTP, SMTP functions overview	★	Medium

2. DNS

Question / Concept	Stars	Importance
DNS query and response encapsulation in UDP	★★	🔥🔥 Very High
Protocols involved in requesting a webpage (DNS + HTTP + ARP)	★★	High
Protocols involved in traceroute (DNS, UDP, ICMP)	★★	Medium
DNS concept and structure (hierarchy, resolver, caching)	★★	🔥🔥 Very High

✅ Sarna Mam — DNS Focus Points:

1. Query/response packet format
2. UDP encapsulation

3. SMTP

Question / Concept	Stars	Importance
SMTP message boundary and comparison with HTTP	★★	🔥🔥 Very High
Explain MIME and its function	★★	Medium
MIME header for JPEG mail	★	Medium

Question / Concept	Stars	Importance
SMTP operation steps (connection, HELO, DATA, QUIT)	★	High

4. UDP

Question / Concept	Stars	Importance
UDP header fields, checksum calculation	★★	High
UDP encapsulation examples (TFTP RRQ/WRQ)	★	Medium
DNS query in UDP	★★	High
Compare TCP vs UDP (reliability, connection)	★★	🔥🔥 Very High

5. TCP

Question / Concept	Stars	Importance
TCP 3-way handshake with ISN example	★★	🔥🔥 Very High
Flow control with window diagram	★★	High
TCP half-close case	★	Medium
TCP segment exchange example (Hello ↔ Hi)	★	Medium
Window & ACK calculation problems	★★★	🔥🔥🔥 Top Priority
Connection establishment and teardown process	★★	High
TCP congestion control, cwnd, rwnd behavior	★★	🔥🔥 Very High